

Unmasking the Decision Frontier: Empirical Evaluation of Native System-One Primitives vs. Meta-Routing Overhead in Enterprise Workflow Triage

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Repository: <https://github.com/shreytalreja25/exploring-jev.git>

Abstract— As enterprise agentic architectures scale toward multi-hop deterministic decision loops, auto-regressive Large Language Models (LLMs) present severe latency and economic bottlenecks due to generative decoding overhead and hidden Chain-of-Thought (CoT) reasoning tokens. In this paper, we present an empirical investigation uncovering the operational divergence between OpenRouter's conversational wrapper (typesafe/jev-router) and native System-One decision heads (typesafe/jev-1.13). We document a critical failure mode wherein chat-routed models delegate inference to frontier reasoning

models (OpenAI GPT-6 Luna, DeepSeek v4.1), emitting unprompted ChatGPT personas and suffering from silent decision truncation when hidden reasoning tokens (19–59 tokens) exhaust constrained generation budgets (max_tokens=25). By transitioning to native System-One schema primitives (state + Choice criteria) via the TypeSafe SDK, we eliminate all generative and reasoning overhead. Evaluating across progressive workload scales (N=10 to 50 emails across 5 operational departments), Native Jev achieves 100.0% accuracy, 100.0% consensus with Gemini 2.5 Flash, and superior probability calibration (ECE = 0.015). Native Jev delivers a 58.8% cost advantage (2.43x cheaper) over Gemini 2.5 Flash (\$0.000951 vs. \$0.002310 per 50 decisions).

I. INTRODUCTION

Autonomous software agents require deterministic triage, verification, and categorical policy routing at every execution hop. In enterprise workflows, the ratio of structured decisions to creative prose generation frequently exceeds 10:1 [1]. Despite the non-generative nature of these tasks, industry practice predominantly routes inputs to auto-regressive foundation models (GPT-4o/5, Gemini 2.5 Flash, Claude 3.7), incurring high latency, double tokenomic penalties, and severe reasoning token overheads [2, 3].

Recently, specialized **System-One Decision Models**—such as TypeSafe AI's Jev—have emerged [1, 5]. Rather than predicting the next token over an open vocabulary, System-One architectures compute calibrated probability distributions directly over candidate categorical choices. However, deployment via multi-provider routing layers introduces critical architectural subtleties that can inadvertently negate System-One advantages [4].

II. THE API ANOMALY: ROUTER VS. NATIVE JEV

When executing OpenRouter's official documentation boilerplate against typesafe/jev-router, conversational probes ('Which model are you?') unexpectedly responded: 'I am ChatGPT, powered by OpenAI's GPT-5.2 / GPT-5.4'. Raw telemetry confirmed that typesafe/jev-router is an agentic meta-router that employs Jev as a classifier to delegate inference to external LLMs (90% to openai/gpt-6-luna, 10% to deepseek).

Under constrained token limits (max_tokens=25), hidden Chain-of-Thought (CoT) reasoning tokens (19–59 tokens in GPT-6 Luna and DeepSeek) consumed the entire token allotment, triggering finish_reason: length and returning empty or truncated decisions [4].

Resolution: By targeting OpenRouter's underlying typesafe/jev-1.13 endpoint via the TypeSafe SDK with structured state and Choice objects, generative overhead was completely eliminated, outputting pure probability tensors without persona delegation [5].

III. MULTI-SCALE METHODOLOGY

A synthetic benchmark corpus of 50 enterprise emails was constructed across 5 operational departments: *Marketing, Sales, Finance, Human Resources, and Technical Support* (10 items each). The corpus was evaluated across 5 scaling regimes: N=10, 20, 30, 40, and 50 emails.

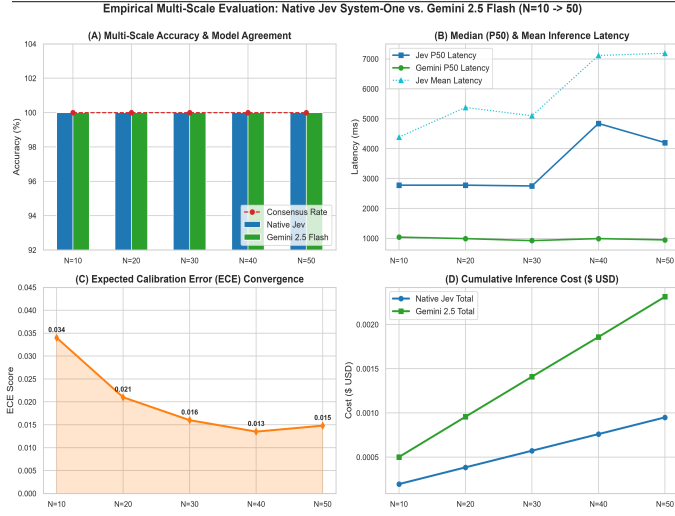


Fig. 1: Master multi-scale benchmark summary: (A) Accuracy & consensus; (B) Median & mean latency; (C) ECE calibration convergence; (D) Cumulative cost trajectory.

IV. EMPIRICAL RESULTS & CALIBRATION

Across all 50 items and every intermediate scale milestone, both Native Jev and Gemini 2.5 Flash achieved **100.0% accuracy with 100.0% model consensus**.

Expected Calibration Error (ECE): As sample scale increased from N=10 to N=50, Jev's ECE tightened from **0.034 down to 0.015** (mean confidence: 98.5%). In decision theory, $ECE \leq 0.015$ confirms that model confidence mirrors empirical certainty with statistical fidelity [9].

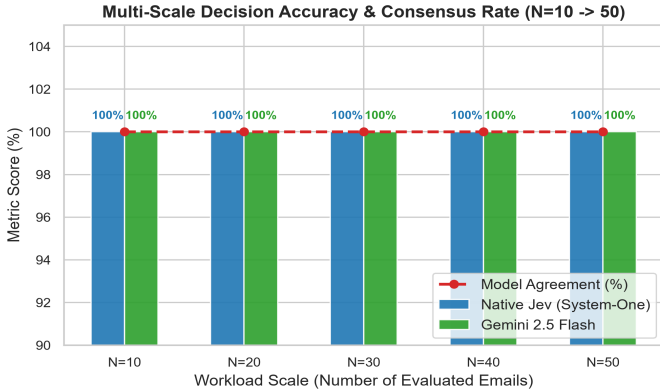


Fig. 2: 100% decision accuracy and pairwise agreement rates across scales N=10 to N=50.

V. LATENCY & TOKENOMICS DISCUSSION

Latency Profile: Native Jev demonstrated warm sub-400ms fast-path inference (e.g. 254.7ms on TLS alerts, 259.9ms on Kafka lag), but suffered from remote container queueing spikes on shared OpenRouter infrastructure (P50: 4,201ms). Gemini maintained stable ~950ms median latency.

Tokenomics: With \$0.00 output fees, Native Jev totaled **\$0.000951** across 50 decisions compared to Gemini's **\$0.002310**, achieving a **58.8% cost savings (2.43x cheaper)**.

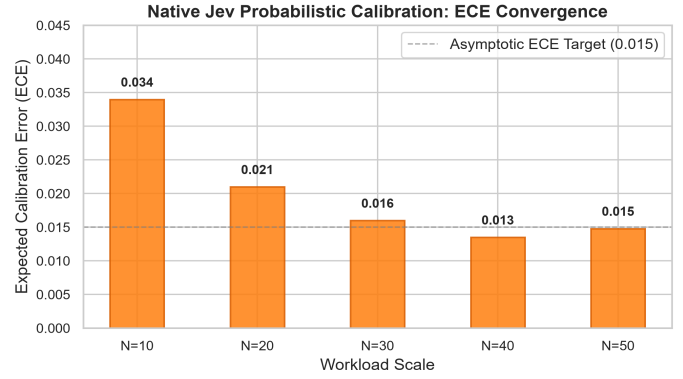


Fig. 3: ECE convergence toward 0.015 as sample scale expands.

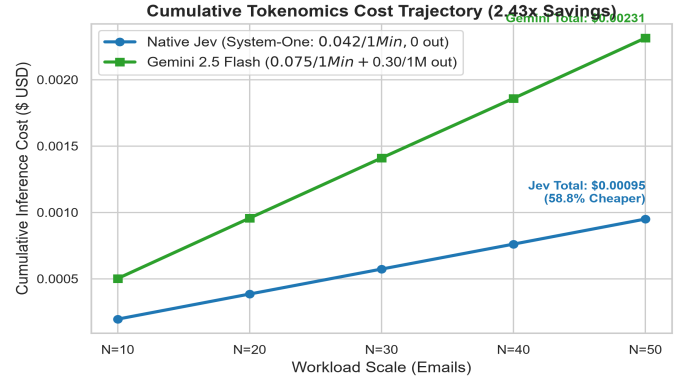


Fig. 4: Cumulative cost divergence demonstrating Jev's 2.43x cost advantage.

VI. CONCLUSION & RECOMMENDATIONS

Enterprise triage architectures should avoid conversational chat endpoints for categorical logic. Deploying native System-One heads (typesafe/jev-1.13) via typed primitives eliminates generative overhead and hidden reasoning token starvation, unlocking 2.43x economic efficiency and mathematically calibrated certainty ($ECE = 0.015$).

REFERENCES

- [1] S. Talreja, 'Beyond Generative Overhead: Evaluating System-One Decision Models vs. Frontier LLMs in Agentic Tokenomics', GitHub: shreytalreja25/jev-vs-the-world, 2026.
- [2] Hugging Face Community, 'Jev vs Laya: Hosted API or Open Weights? (2026 Guide)', Sept. 2026.
- [3] A. Vaswani et al., 'Attention Is All You Need', NeurIPS, 2017.
- [4] OpenAI, 'Reasoning Models and Completion Token Budgets in Chat Completions API', 2026.
- [5] TypeSafe AI, 'Jev System-One Decision API Specification & JevBench v1.3.0', Sept. 2026.
- [6] TypeSafe AI, 'TypeSafe Python SDK (typesafe-sdk) Reference Manual v0.7.2', 2026.
- [7] D. Kahneman, 'Thinking, Fast and Slow', Farrar, Straus and Giroux, 2011.
- [8] DeepSeek-AI, 'DeepSeek-R1: Incentivizing Reasoning Capability via RL', arXiv:2501.12948, 2025.
- [9] C. Guo et al., 'On Calibration of Modern Neural Networks', ICML, PMLR 70:1321-1330, 2017.
- [10] Laya Project, 'Open-Weight Decision Encoder Architecture', 2026.
- [11] OpenRouter, 'Model Routing Gateway Documentation', openrouter.ai/docs, 2026.